Supplementary Information

Supplementary Methods S1. Reproduction environment and source files

The code and test suite are maintained at https://github.com/omixcore/phytoscope. Before submission, replace the moving branch URL with a permanent release/archive DOI and record the exact commit hash used for the article. Species accessions, annotation versions, URLs, and integrity checks are defined in the repository manifests. The three non-model NCBI GFF3 downloads are checksum-verified at fetch time. The rice-bean RNA FASTA and GFF3 are pinned by SHA-256, and six paired FASTQ files are pinned by byte size and MD5 in `src/test/resources/non-model/vigna-umbellata/public-expression-manifest.json`.

Reproduction commands

Use Java 21 for the application tests. Public reference files are fetched into a generated directory and are not required to be checked into the source repository.

```
python tools/fetch_vigna_reference.py --directory target/non-model/vigna-umbellata
python tools/fetch_additional_nonmodel_refs.py --directory target/additional-nonmodel
python tools/build_nonmodel_sequence_gold.py
./mvnw -B test
```

The dedicated rice-bean integration test is:

```
./mvnw -B -Dtest=NonModelVignaReferenceTest \
  -Dphytoscope.vignaReferenceDir=target/non-model/vigna-umbellata test
```

For the archived public expression calculation, use Salmon 2.8.0 and the exact RNA FASTA and GFF3 identified in the manifest:

```
python tools/requantify_public_expression.py \
  --rna-fasta target/non-model/vigna-umbellata/GCF_029865625.1_rna.fna \
  --gff target/non-model/vigna-umbellata/GCF_029865625.1_genomic.gff.gz \
  --reads-dir /path/to/fastq \
  --salmon /path/to/salmon \
  --output target/non-model/vigna-umbellata/expression \
  --threads 4 --fetch-reads --temporary-reads
```

The script validates fetched read size and MD5, transcript-to-gene mapping, matrix parsing, and output hashes. The checked-in `public-expression.tsv` is the resulting import fixture. It is explicitly not a differential-expression result.

Table S1. Species and data coverage

Taxon	Assembly / annotation	Data evaluated	Test scope
Arabidopsis thaliana	TAIR10/Araport11; Ensembl Plants release 62	Full annotation, chromosome FASTA, selected expression and protein examples	Full recognized-feature tuple audits; promoter, CDS QC, homology, expression arithmetic, domain mapping; app regressions
Oryza sativa	IRGSP-1.0; Ensembl Plants release 62	Full annotation and source excerpt	Full tuple audit; multi-chromosome parser comparison
Zea mays	B73 NAM5; Ensembl Plants release 62	Full annotation and source excerpt	Full tuple audit; multi-chromosome parser comparison
Glycine max	v2.1; Ensembl Plants release 62	Full annotation and source excerpt	Full tuple audit; multi-chromosome parser comparison
Vigna umbellata	GCF_029865625.1, Vumbellata_1.0; GCF_029865625.1-RS_2026_09	Full GFF3 and GTF; matching chromosome FASTA and RNA FASTA; three public read libraries	Full imports; query/structure; independent selected-sequence gold; Salmon re-quantification and exact ID reconciliation
Andrographis paniculata	GCF_009805555.1; Annotation Release 100	Full GFF3; selected scaffold FASTA	Full GFF3 import/search; selected-locus structure and sequence reconstruction
Hibiscus syriacus	GCF_006381635.1; Annotation Release 100	Full GFF3; selected scaffold FASTA	Full GFF3 import/search; selected-locus structure and sequence reconstruction

Table S2. Full annotation tuple inventories

Inventory	Tuples	Missing	Extra
Arabidopsis bundled annotation	811,827	0	0
Arabidopsis full release 62	791,557	0	0
Maize source excerpt	377	0	0
Maize full annotation	1,161,691	0	0
Rice source excerpt	347	0	0
Rice full annotation	550,921	0	0
Soybean source excerpt	839	0	0
Soybean full annotation	1,324,199	0	0
Total	4,641,758	0	0

Tuple fields include recognized gene, transcript, exon, CDS, and UTR type, explicit parent relation, sequence ID, strand, and exact coordinates. The validator documents ignored or unsupported source feature types separately.

Table S3. Public expression quantification inputs and output

Field	Value
BioProject	PRJNA765494
Genotype / developmental stage	IC426787; 5 days post anthesis
Runs	SRR16122599, SRR16122608, SRR16122607
Assembly / cultivar	GCF_029865625.1 / Vumbellata_1.0 / VRB3
Annotation release	GCF_029865625.1-RS_2026_09
Quantifier	Salmon 2.8.0; selective alignment; k-mer length 31
Reference transcript count	54,178; RefSeq RNA FASTA SHA-256 in public-expression-manifest.json
Gene-map rows / matrix rows	36,754 map entries / 34,914 quantified gene rows
Exact annotation ID matches	34,914 of 34,914 matrix IDs; 0 unmatched
Missing / invalid cells	0 / 0
TPM column sums	1,000,000.000034; 999,999.999897; 999,999.999951
Interpretive limit	Genotype differs from reference cultivar; import and identifier compatibility only; no differential-expression inference

Table S4. Additional module validation panels

Function	Independent reference	Observed result	Scope limit
Promoter extraction	Direct 1-based inclusive FASTA slicing and reverse complement; bedtools 2.31.1	8/8 exact sequences, 2,200 bp each, both strands represented	Eight loci on one Arabidopsis chromosome
CDS sequence quality	Independent base-count/frame/start/stop oracle	Four RBCS CDS models (411, 561, 546, 546 nt) matched; no ambiguous bases flagged	Four models from one family

Function	Independent reference	Observed result	Scope limit
Protein similarity	Test-side Smith–Waterman and BLASTP 2.12.0+	All 16 within-panel alignment metric checks; 12/12 non-self RBCS and 8/8 negative pair classifications agree; 20/20 in the cross-species mature-RBCS panel	Small one-family smoke test; not orthology, sensitivity, or specificity evaluation
Expression grouping	Expression Atlas E-MTAB-7978 source values	Selected means/medians reproduced; measured zeros retained	Single-assay group arithmetic, not differential expression
Domain mapping	Independent Python reconstruction	16 domain/deletion coordinate records across 13 genes matched	Existing annotated models; not a curated multi-species accuracy set
Project persistence	JavaFX regression actions	Save, replace, restore, locus state and outputs verified in CI	Software behavior, not usability study

Supplementary Note S1. Data required from users and non-inferable properties

Sequence extraction requires a genome FASTA matching the annotation assembly. Coordinates, feature names, and parent relationships are those recorded in the imported annotation; they are not independent confirmation of model quality. Expression display requires a matrix and, for biological interpretation, a clear unit/normalization, sample identity, condition, replicate design, and source metadata. The software cannot infer functional roles, orthology, regulatory effects, differential expression, unannotated isoforms, or biological absence from missing feature rows. Unprovided annotation fields remain unknown; derived intervals are labeled as inferred.

Supplementary Note S2. Fair-comparison boundary

The completed `gffutils==0.13` comparison is a scripted annotation hierarchy extraction baseline only. Manual file handling and JBrowse/IGV interactions were not run as matched end-to-end tasks. No participant study was conducted. Thus the manuscript reports no speed ranking, usability advantage, user time savings, reduced error rate, or user-value effect. A future comparison should freeze the same species, assembly, files, locus list, output specification, and independent gold standard, then record commands and applications, intermediate files, provenance fields, and outputs across counterbalanced runs.